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Scattered palaeomagnetic directions acquired during dioritization and stoping of the diorite-metagabbro complex, Jersey, C.I.

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SUMMARY: Scattered, but stable NRM (natural remanent magnetization) in the Jersey diorite-metagabbro complex, probably acquired during the formation of the complex, can be ascribed to 2 main geological processes: (*i*) stoping of the late Precambrian or early Cambrian diorites and metagabbros by the Jersey granites, which caused pre-stoping NRM to be dispersed by block rotation and subsidence in the granite melt; this was the largest contributor to palaeomagnetic scatter; (*ii*) hybridization of the parent gabbro during granite intrusion, which added a dioritic generation of magnetite and its NRM to the NRM carried by the gabbro, and probably also caused variable, partial remagnetization of the gabbroic NRM before stoping.

Palaeomagnetic results suggest that the complex NRM of the Jersey diorite-metagabbro complex was not acquired as a VRM (viscous remanent magnetization) in a recent geomagnetic field. Rather, the explanation of the unusual scatter of stable NRM directions in these rocks is probably to be sought in the well-documented sequence of geological events that governed the formation of the diorite-metagabbro complex (Wells & Wooldridge 1931; Bishop 1954; Wells & Bishop 1955; Key 1974).

This complex outcrops on the N and SE coasts of Jersey (Fig. 1). On the S coast it consists of rhythmically layered sheets which vary upward from fine-grained meladiorite at the base, through medium-grained mesodiorite to coarse-grained appinite, which may have originated from the diorite by *in situ* melting (Key 1974). The dioritic mineralogy was imposed during intrusion and hybridization of a layered gabbroic intrusion by the Jersey granites (Key 1974) during the Cambrian. The granites were probably emplaced mainly by cauldron subsidence and stoping of dioritic and metagabbroic blocks.

Palaeomagnetic collections were made within blocks sampled at 8 localities in the SE diorite along the coast from Elizabeth Castle to Le Nez (Fig. 1). The section at Le Nez (LZ1 to LZ4) was intensively sampled owing to good continuous exposure and control on the attitude of fine-scale igneous layering. The NW diorite was sampled at 2 sites in the large quarry at Ronez and one site at Sorel Point on the N coast. The attitude of fine-scale igneous layering in the NW diorite was determined at Sorel Point.

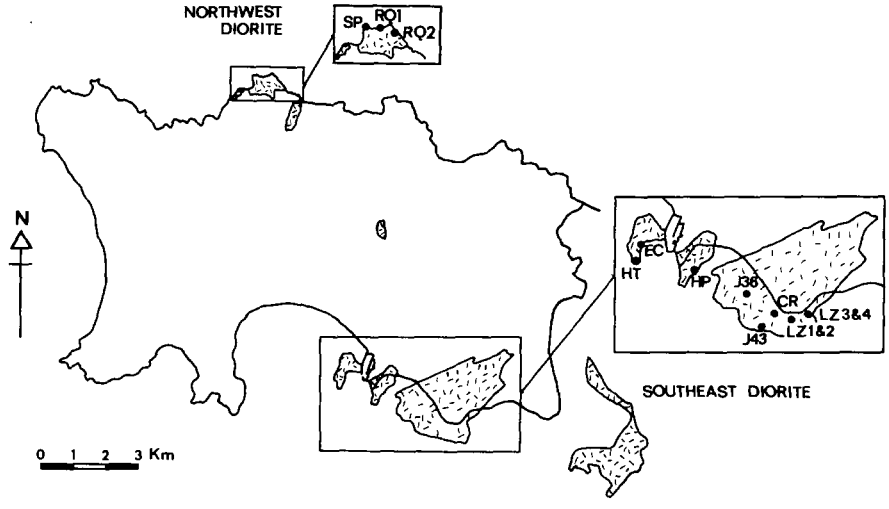


FIG. 1. Sketch map of Jersey (Long. 358° E, Lat. 49° N) showing palaeomagnetic sampling localities in the NW and SE diorites (patterned outcrop area). Localities referred to in the text: Sorel Point (SP); Ronez Quarry (RQ); Elizabeth Castle (EC); and Le Nez (LZ).

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Nature of the NRM

The antiquity of the metagabbroic and dioritic NRM is reflected in the results of thermal demagnetization (Fig. 2) as a significant, although variable, portion of total NRM residing in the high blocking temperature range 400–600° C. Equally, resistance to af (alternating field) demagnetization (Fig. 3) is high, and specimens typically retain up to one-tenth of their original NRM intensity after demagnetization in peak fields of 100 mT (Fig. 3). The directional variation of NRM during both af and thermal demagnetization, however, is highly variable between stoped blocks, and less variable within blocks. All types of directional variability are represented, from single end-points (e.g. CR1*B*, Fig. 3*b*), to coherent, near-planar movements of resultant NRM (e.g. Fig. 2*b* & CR2 of Fig. 3*b*) to non-planar, though non-erratic movements (e.g. Fig. 2*a*). Although some variability in these directional responses within blocks is general, there are many examples of specimens within blocks with very similar resultant NRM trends (e.g. Figs 2*b* & 3*b*).

Contact tests of the SE diorite against a narrow dolerite dyke and an appinite further support the stability and antiquity of NRM in the diorite-metagabbro complex. Specimens of diorite collected from within 12 cm of the dolerite contact have very similar directions of total NRM to the mean total NRM direction of the dyke, while specimens further from the contact have directions intermediate between those in the immediate contact zone and those in the diorite remote from the dyke (Fig. 4*a*). After isolation of stable NRM, the diorite mean direction is little changed. Both thermal and af demagnetization, however, cause the resultant NRM of the contact diorite to rotate away from the dyke direction.

Similar rotation of the resultant NRM of the contact diorite away from the NRM direction of the younger intrusive characterizes the appinite contact test (Fig. 4*b*). In this example, thermal demagnetization caused contact diorite directions to rotate close to a plane from the stable appinite direction to the direction of stable NRM for the diorite, independently determined by af demagnetization.

Although the interpretation of contact tests of NRM antiquity is seldom completely unambiguous, these results for SE Jersey strongly support partial remagnetization of a dioritic or gabbroic NRM, older than the age of emplacement of dolerite (no younger than mid-Palaeozoic) and appinite (Cambrian).

The 17 mean directions of stable NRM for blocks in the SE diorite, and the single mean direction of the 3 NW diorite sites are scattered (Fig. 5*a*, Table 1) and random at the 95% probability level (Watson 1956). A two-tier analysis (Watson & Irving 1957) indicates that the between-block scatter is significant at the

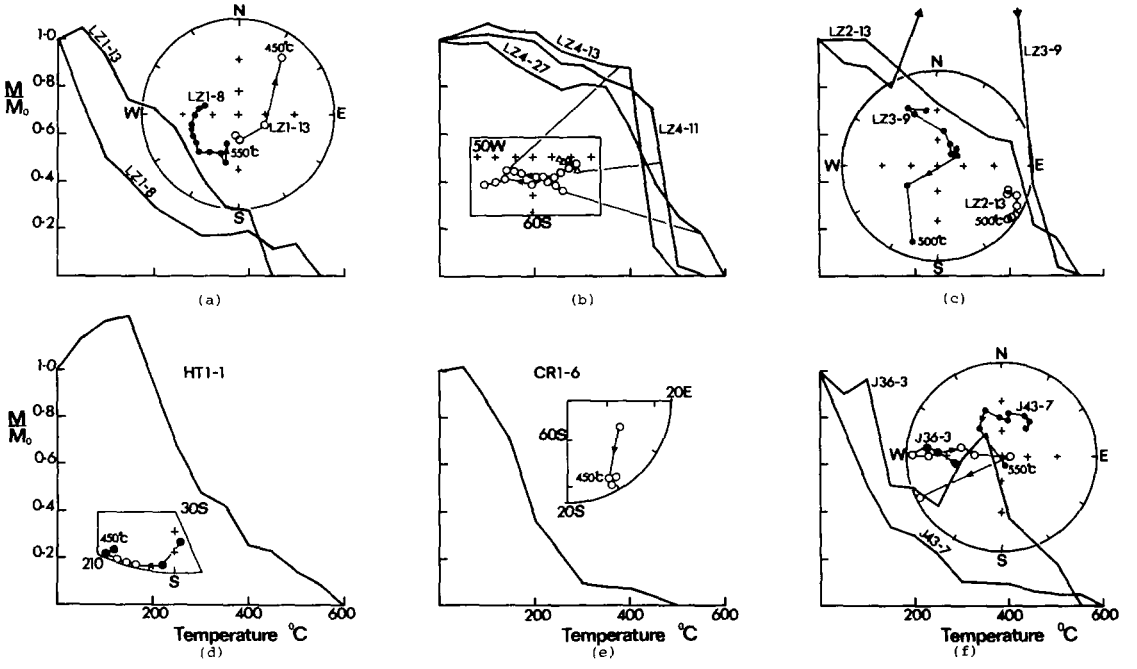


FIG. 2. Thermal demagnetization of the NRM of selected specimens in the Jersey diorites. The corresponding directional changes are shown (closed symbols: downward directions; open symbols: upward directions; stereographic projections).

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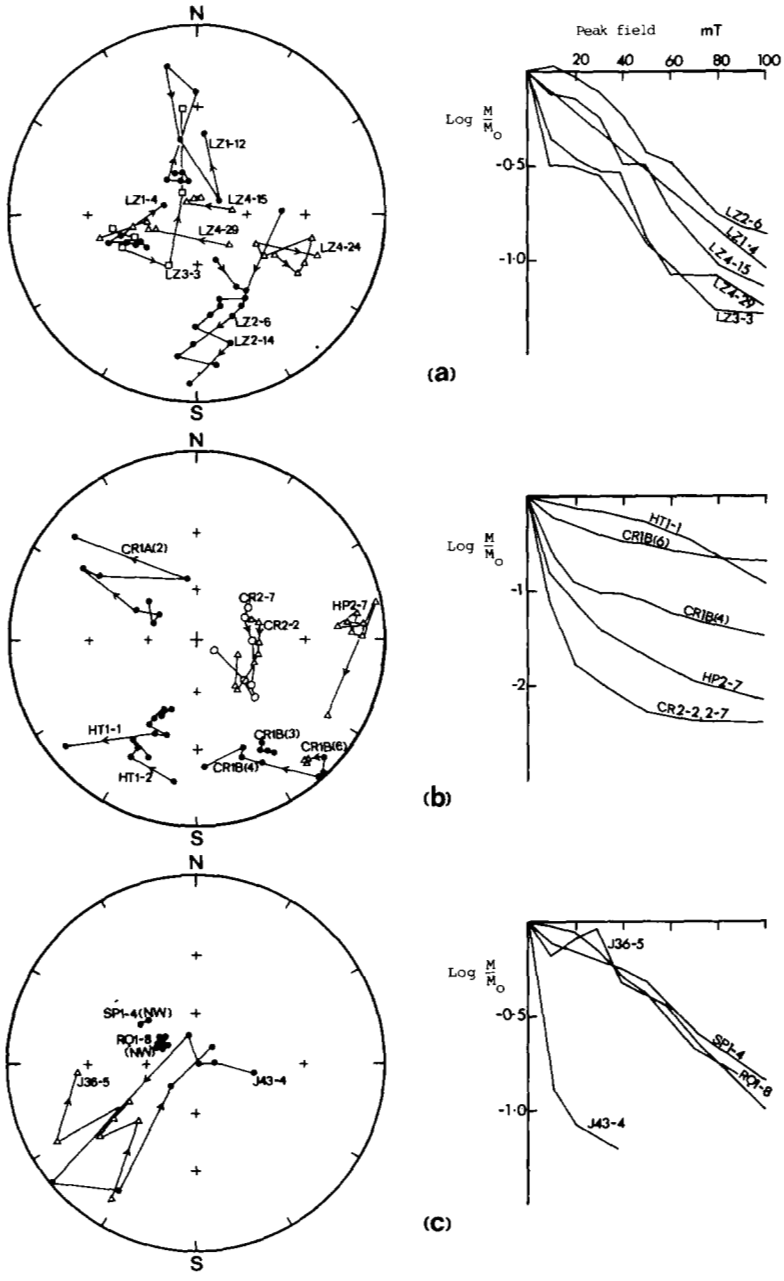


Fig. 3. Af demagnetization characteristics of the Jersey diorites: (*a*) Sites at Le Nez (SE Jersey); (*b*) Other SE diorite sites; (*c*) NW diorite sites (RQ1 & SP) and sites J43 and J36 (SE diorite). Symbols and projections as in Fig. 2; M is intensity of NRM. Different symbols (circles, squares, triangles) are used simply to distinguish different specimens.

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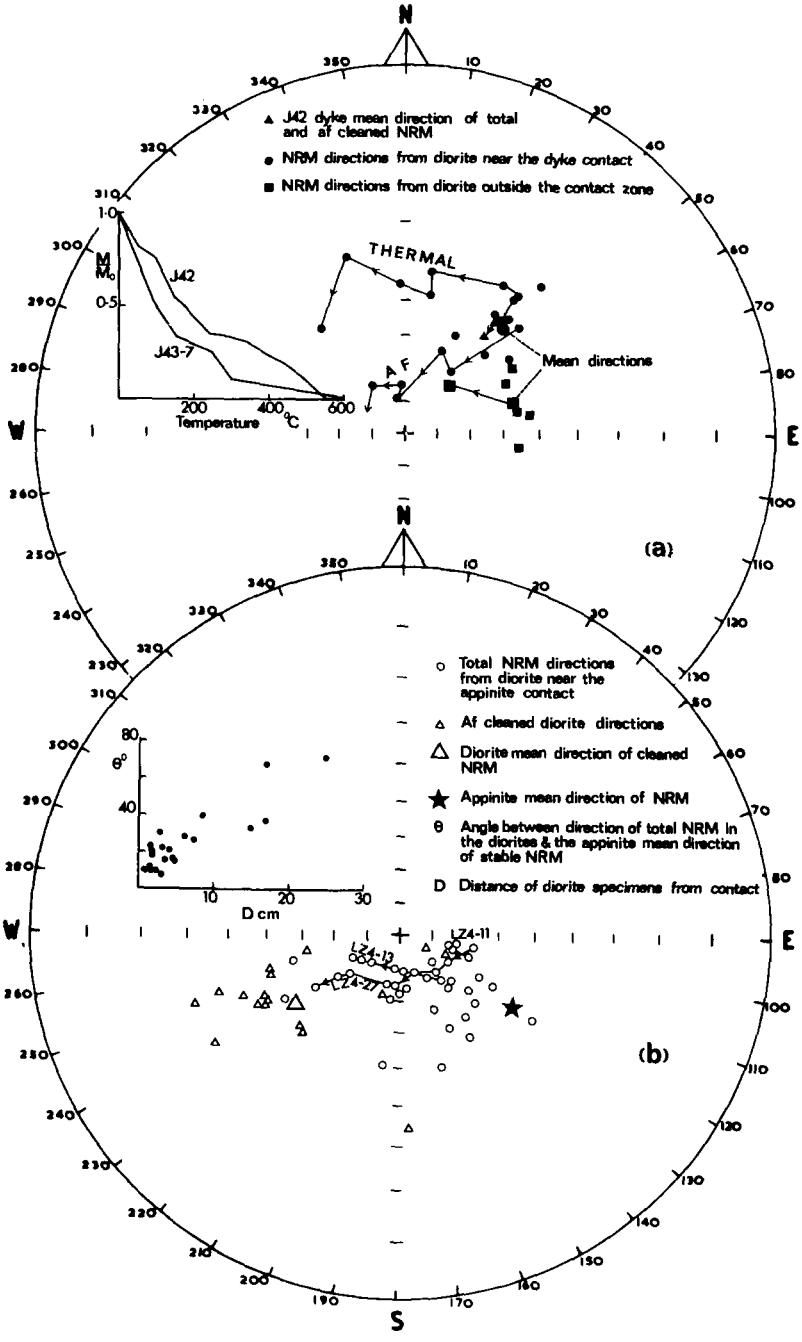


FIG. 4. Contact tests of NRM in the SE diorite. (*a*) Comparison of blocking temperature spectra of dyke J42 with the spectra of diorite (J43) in its contact zone. Mean directions of dyke and diorite in and outside the contact zone before and after af cleaning, and the directional trends of diorite contact NRM on demagnetization, are also shown (all directions downward). (*b*) Comparison of LZ4 appinite mean NRM direction after af cleaning (star) with NRM directions of the adjacent diorite before and after af cleaning. Typical thermal demagnetization trends for the diorite are indicated (all directions upward). The relationship between degree of remagnetization (θ) of diorite specimens, and their distance from the appinite contact is also shown. Stereographic projections.

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TABLE 1: *Mean directions of NRM* (in situ) *of the Jersey diorite-metagabbro complex*

	N	D	I	R	k	α_{95}	*Palaeomagnetic pole* Long.	Lat.	(dψ, dx)
NW diorite	3 (sites)	325°	60°	2.99	170	9	264° E	64° N	(11, 14)
SE and NW diorite combined	18 (sites and subsites)	174°	32°	4.34	1.2	67			
Two-tier analysis: (SE and NW diorite combined)	$\omega = 7.3$	$\beta = 1.5$		$\alpha_{95} = 27$		$F_{34,246} = 37.5$			

99% probability level ($F_{34,246} = 37.5$), and it emphasizes the observation from demagnetization results that the within-block precision is also low ($\omega = 7.3$).

Two main types of scatter therefore contribute to the overall dispersion of stable NRM directions within the Jersey diorite-metagabbro complex: between-block (the largest), and within-block scatter.

Origin of the between-block scatter

The stoping relationships between granite and diorite suggest that the layering of the diorite sheets was continuous before granite intrusion (Wells & Wooldridge 1931), and that separate stoped diorite blocks subsided and rotated into their present attitudes as indicated by the variable orientation of layering. A fine-scale layering—probably primary cumulate layering from the parent gabbro (Key 1974)—is parallel to the rhythmic layering of the diorites, and has been intruded perpendicularly by curious cylindrical 'pipes' of appinite. Bishop (1963) interpreted these as diapiric structures that were intruded vertically, perpendicular to the cumulus layering before stoping. This supports the indications from stoping relationships that the diorites and metagabbro subsided and rotated to their present attitude in the intruding granite.

Although there is no means of controlling the original azimuth of the stoped blocks, most blocks have mean NRM directions with fairly consistent inclinations of $0 \pm 30°$ relative to the igneous layering (the primitive of Fig. 5*b*). Inclinations that are downward relative to the igneous layering (in blocks LZ4 and CR2, Fig. 5*b*) would be equivalent to the upward inclinations if the dioritic blocks were inverted during stoping. Because the layering is known from field relationships to have been initially near-horizontal, this consistency of inclination implies that the bulk of NRM dates from before subsidence and rotation. The stable NRM of the NW diorite, however, is inclined at 47° to the igneous layering, and that of block LZ2 at 60° (Fig. 5*b*); if the between-block scatter is indeed due to stoping, stable NRM must have been acquired in these rocks either before or after the pre-stoping NRM of most blocks. The metagabbroic mineral assemblage of the NW diorite suggests that it has been little altered from the parent gabbro (Key 1974). Accordingly, the steeper direction of the NW diorite NRM probably reflects an older magnetization than in most of the SE diorite blocks.

The association of different mean NRM directions with small, separately stoped diorite blocks within a single sampling site in SE Jersey strongly supports scattering of an NRM acquired before stoping by block rotation and subsidence. This is analogous to the scattered NRM of clasts in the conglomerate test (Graham 1949).

Although the scatter of stable NRM directions in the diorites is best explained by scattering during block rotation after stoping, not all blocks could be corrected to their prestoping positions to test rigorously the prediction of similarly inclined NRM directions. It might therefore be possible to argue that the observed similar inclinations of most dip-corrected NRM directions is fortuitous. Alternative, though less plausible, explanations include: (1) Differential remagnetization of an original gabbroic NRM during dioritization. This probably explains much of the within-block scatter, and although it may have contributed to some of the between-block scatter, comparison of within-block and between-block precisions (7.3:1.5) implies that its contribution has not been great. (2) A non-dipolar geomagnetic field during the early Cambrian. There is no record of non-dipole effects in the Cambrian, and even if the field was non-dipolar, it would be surprising if the field directions were so variable over such a small region as the Jersey diorite complex. (3) Protracted cooling of the diorite-metagabbro complex during a period of rapid, erratic apparent polar wander. This interpretation requires that sites with variable blocking temperatures recorded different field directions. It is inconsistent, however, with the oblique NRM directions from adjacent diorite blocks which might be expected to have cooled together. Furthermore, contrary to the predictions of this interpretation, the stable directional

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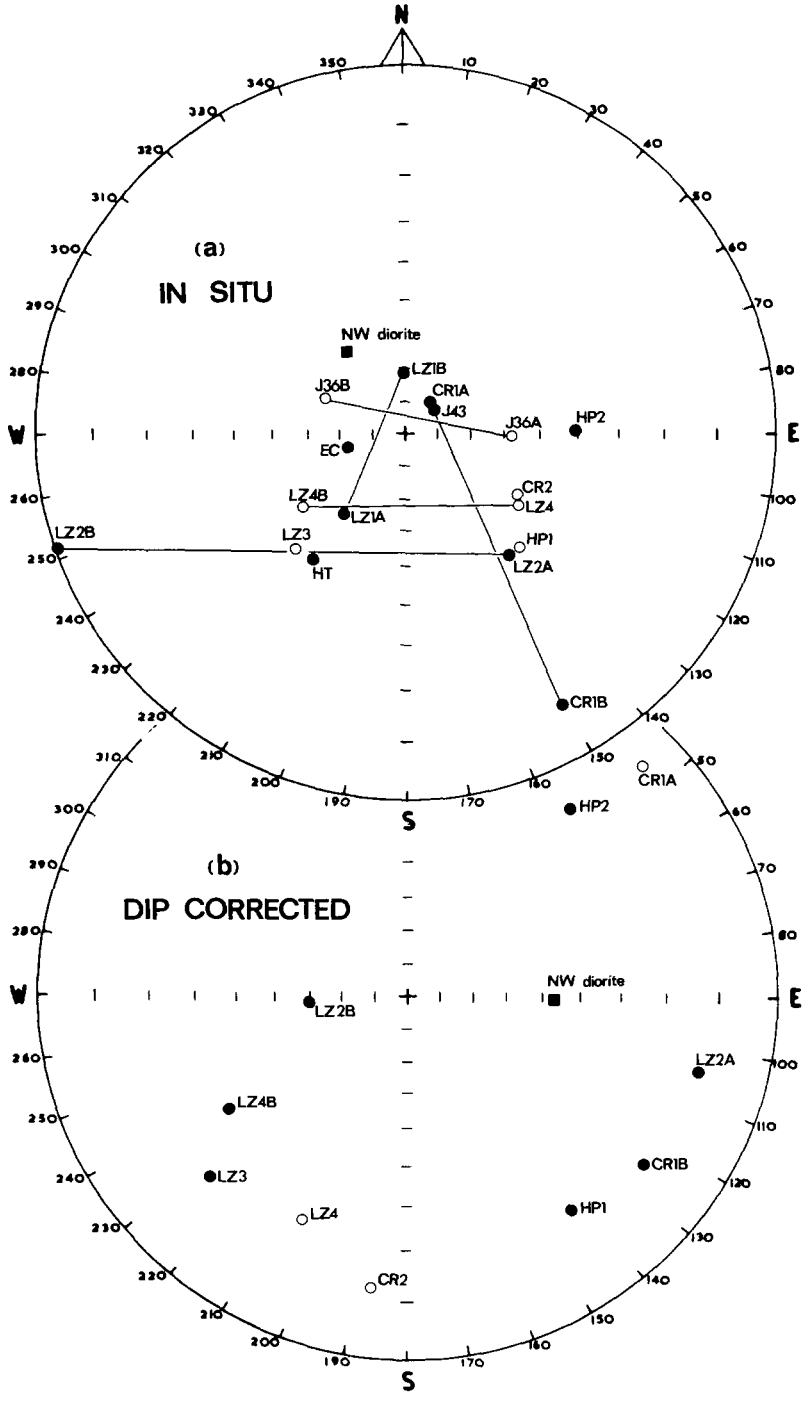


FIG. 5. (*a*) *In situ* block mean directions of NRM for the Jersey diorite-metagabbro complex after optimum af cleaning. Tie lines connect sampling localities within the same stoped block. (*b*) Block mean directions after af cleaning, corrected for the dip of igneous layering. Closed symbols: downward directions; open symbols: upward directions. Stereographic projections.

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changes associated with distributed blocking temperature spectra show no relation to block mean NRM directions.

Origin of the within-block scatter

Microprobe and optical results (Duff 1978) strongly suggest that the within-block scatter of NRM directions and related variable demagnetization trends (Figs 2 & 3) reflect the highly variable overprinting of an original gabbroic NRM during hybridization of the gabbro (dioritization) by the Jersey granites. The resulting multicomponent NRM may have originated in either or both of the following ways: (*i*) variable partial remagnetization of primary gabbroic NRM; (*ii*) acquisition by magnetites grown during dioritization of a dioritic NRM in addition to that already present in the primary gabbroic titanomagnetites.

Typically, the original gabbroic assemblage has been altered to a dioritic assemblage at temperatures of the upper amphibolite or granulite facies (Key 1974). At these temperatures (>500° C) only a small amount of the original remanence may survive in grains with high blocking temperatures (Pullaiah *et al.* 1975), so it is likely that the original gabbro was entirely or partially remagnetized during hybridization by the granite melt. The intricate field relationships in the Jersey diorite-metagabbro complex imply that this process was likely to have been highly variable or heterogeneous on a within-block scale.

Two types of magnetite (*sensu lato*) are distinguished in the Jersey diorites. Type 1 form large, skeletal grains with ilmenite or haematite lamellae in the host {111} cleavage. Microprobe analysis yields titaniferous compositions for these magnetites: $Fe^{3+}_{0.86}$ $Fe^{2+}_{1.52}$ ($Ti_{0.56}$ $V_{0.01}$ $Mn_{0.02}$ $Co_{0.03}$) O_4. In contrast, type 2 magnetites form small, homogeneous grains and are more abundant than the type 1 grains. Microprobe analysis confirms their homogeneity and indicates compositions very close to pure end-member magnetite. These type 2 magnetite compositions are out of place in gabbroic rocks in which the molar percent ulvöspinel in magnetite generally ranges from about 15–60% (Buddington & Lindsley 1964). The fresh, homogeneous titanium-poor nature of these magnetites, however, is consistent with crystallization during dioritization and is comparable to the homogeneous, titanium-poor compositions of magnetites in the Jersey granites (Duff 1978). The composition of the skeletal type 1 magnetites, on the other hand, suggests that they may be relict from the parent gabbro.

Variable proportions of type 1 and type 2 magnetites in the Jersey diorites, carrying respectively gabbroic and dioritic NRM, could therefore perhaps, in addition to a multi-component NRM associated with variable partial remagnetization of the gabbroic NRM, account for the within-block scatter of the Jersey diorites.

Magnetization history in relation to the geological evolution of the diorite-metagabbro complex

Palaeomagnetic and geological results together support the following history:

(1) Before intrusion and stoping by the granites there existed a continuous, near-horizontal layered gabbro over much of Jersey, which probably had a uniform directon of NRM carried by titanomagnetite (Fig. 6*a*).

(2) In the Cambrian or Ordovician, granites were intruded at high structural levels into the base of this gabbroic pile. During the ensuing hybridization of gabbro by granite melt, temperatures reached at least amphibolite facies grade, and most original gabbroic phases were converted to a dioritic assemblage (Key 1974). A variable, multi-component NRM was imposed on the diorite by variable partial remagnetization of the blocking temperature spectra of gabbroic titanomagnetites during dioritization, or by addition to the older gabbroic or dioritic NRM of a separate component of NRM carried by magnetite, which crystallized during dioritization (Fig. 6*b*).

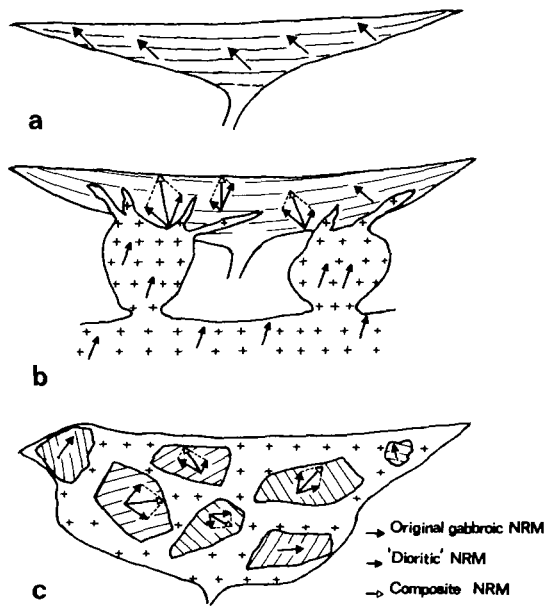


FIG. 6. Schematic representation of the acquisition of variable, multicomponent NRM during dioritization and its scattering during block stoping by the Jersey granites. (*a*) Initial gabbroic intrusion and acquisition of gabbroic NRM. (*b*) Granite intrusion; dioritization of parent gabbro and variable acquisition of a 'dioritic' component of NRM in addition to the initial gabbroic NRM. (*c*) Cauldron subsidence and rotation of stoped diorite blocks into granite melt with scattering of multicomponent NRM directions.

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(3) After further higher level intrusion by the granite, subsidence of the diorite sheets caused large blocks—perhaps as large as the dioritic region from Ronez to Sorel Point—to subside and rotate into the granite melt. Further intrusion was achieved by brittle fracture, whereby smaller blocks of diorite were stoped into the granite. At both scales of stoping, the NRM directions acquired before stoping, whether the original gabbroic NRM (NW diorite?) or a subsequent multi-component NRM imposed during dioritization, were scattered by rotation and subsidence in the granite melt (Fig. 6*c*).

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